

David Sierra Sosa

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PROFESSIONAL SUMMARY

Frontend Developer focused on the React/Next.js ecosystem with hands-on experience integrating Artificial Intelligence workflows and vector databases. Oriented towards creating scalable architectures and solving domain-specific problems using modern development tools.

EXPERIENCE

Frontend Development Intern

May 2025 – Aug. 2025

Genotipo

Cancun, Quintana Roo

- Optimized and maintained user interfaces in existing web applications, resolving UI/UX issues and ensuring full cross-device responsiveness across mobile, tablet, and desktop using PHP (CodeIgniter) and Tailwind CSS.
- Structured and developed base views for the “Arte Gasolineras” project, integrating responsive design and scalable code within an MVC architecture to streamline engineering team handoff.
- Managed version control and code collaboration through Bitbucket, implementing the Gitflow methodology to ensure development branch quality and smooth deployment pipelines.

FEATURED PROJECTS

HEXACORE - AI Recommendation System [\[GitHub\]](#) | *React 19, pgvector, Gemini API, PostgreSQL*

- Designed and implemented a RAG (Retrieval-Augmented Generation) architecture combining vector databases for semantic searches and structured relational storage for domain rule management.
- Developed a validation system (agentic loop) with automated retry mechanisms to evaluate the viability of LLM responses, eliminating hallucinations and ensuring structured JSON outputs.

Gestor Agrícola (Agricultural Dashboard) [\[Frontend\]](#) [\[Backend\]](#) | *React, Firebase, Mapbox GL, IoT*

- Developed a full-spectrum monitoring platform integrating interactive dashboards with geospatial maps for real-time data stream visualization.
- Designed secure authentication workflows and persistent real-time database management using Firebase and JWT.

Barbershop Management System (Zotz Barber) [\[GitHub\]](#) | *React Router v6, Google Maps API, Node.js*

- Built a comprehensive web/mobile system for appointment and client scheduling in a real-world business case, optimizing final user experience through dynamic routing handlers.
- Integrated interactive location services by directly consuming the Google Maps API to optimize establishment geolocation.

AI Dashboard - Academic Impact [\[GitHub\]](#) | *Next.js, Tailwind CSS, Recharts, PapaParse*

- Developed an analytical dashboard based on Next.js to process, parse, and graph complex structured data regarding the impact of AI technologies within educational environments.
- Implemented interactive visualization components and dynamic data tracking using Recharts to streamline visual analysis.

TECHNICAL SKILLS

Frontend: React (v19), Next.js, Tailwind CSS, React Native, HTML/CSS/JavaScript, TypeScript

Backend & Databases: PHP (CodeIgniter), Node.js, NestJS, PostgreSQL, MySQL, Firebase

AI & Tools: Gemini API, Cursor Pro, RAG integrations, pgvector, GitHub, Bitbucket

Languages: Spanish (Native), English (B1 - TOEFL iBT Certified)

EDUCATION

Universidad Tecnológica de Cancún

Bachelor of Science in Software Development and Management

Cancun, Quintana Roo

2025 – 2027 (Expected)

Universidad Tecnológica de Cancún

Associate Degree in Information Technology (Software Development)

Cancun, Quintana Roo

2023 – 2025